

CITY UNIVERSITY OF HONG KONG

Application Form for Huawei Seed Grant

1 Name of the applicant, position and home Department, and the date of joining the University

- Name of applicant: Chen MA (00473705)
- Position and affiliated department: Assistant Professor, Department of Computer Science
- Date of joining CityU: August 25, 2021
- Duration of the project: 2 years

2 Title of the Proposed Research

- English Project Title: Post-training Data Valuation for Large Language Models
- Chinese Project Title: 大语言模型的后训练数据估值

3 Abstract of the Project

Large Language Models (LLMs) have become the backbone of contemporary natural-language applications, yet their post-training stage—where instruction–response (I–R) pairs are used to align model behaviour with human intent—remains an empirical process highly sensitive to data quality. Existing fine-tuning pipelines either treat each I–R pair as an inseparable unit or focus on one side in isolation, ignoring the *structural asymmetry* between instructions and responses. This blind spot leads to inefficient data evaluation, expensive trial-and-error retraining, and a lack of transparent attribution regarding why certain samples matter.

This project aims to build a **structure-aware, efficient, and interpretable data-valuation framework** for post-training LLMs. We will: 1) design perturbation- and reconstruction-based experiments to *quantitatively profile* the marginal contributions of instructions and responses; 2) develop a *composite valuator* that integrates instruction robustness, response information value, and their interaction strength while maintaining global diversity; and 3) devise a *retraining-free proxy evaluator* that predicts strategy performance via low-rank collaborative filtering and explains sample importance through structure-aware Shapley attribution.

The expected outcome is a principled toolkit that reduces fine-tuning cost by prioritising the most valuable and complementary data, improves downstream task performance, and offers actionable insights into how each sample drives model generalisation. The framework is model-agnostic and readily pluggable into existing instruction-tuning workflows, thereby advancing both the efficiency and trustworthiness of LLM post-training.

4 Key Objectives

This project builds a structure-aware, efficient, and interpretable data-valuation framework for instruction-tuned LLMs. Our objectives are:

- Quantitatively analyse the functional asymmetry between instructions and responses through single-sample structural sensitivity analysis, clarifying their distinct roles and marginal contributions during fine-tuning.
- Devise a composite valuation mechanism that captures both intra-pair interaction and inter-sample relationships, increasing the diversity, complementarity, and representativeness of selected data.
- Create a retraining-free, interpretable evaluation pipeline that assesses data-selection strategies via proxy modelling and structure-aware Shapley attribution, providing rapid feedback and transparent rationale.

5 Research Methodology

5.1 Background of Research

With the rapid advancement of artificial intelligence, large language models (LLMs) have become foundational technologies supporting a wide range of natural language applications, including open-domain question answering and programming assistance. As a key paradigm for adapting pretrained LLMs, post-training involves optimizing models using task-specific datasets composed of instruction-response pairs, effectively aligning model behavior with human intent. This approach has significantly improved the controllability, practicality, and task adaptability of LLMs, and has become an indispensable component in nearly all mainstream LLM pipelines.

Existing research has shown that the effectiveness of fine-tuning heavily depends on the quality and structural characteristics of the data, with some training samples contributing significantly more to model performance than others [1, 2]. This has driven increasing interest in more efficient and accurate data valuation and selection strategies. However, most current data valuation methods treat instruction-response pairs as inseparable units [3, 4], overlooking the fact that instructions and responses serve distinct functional roles during training [5]. For instance, while paradigms like Self-Instruct [6] have demonstrated that high-quality instructions can enhance model generalization, recent independent work has shown promising results in developing methodologies to separately evaluate and select instructions [7]. On the other hand, studies on response-tuning [8] have shown that fine-tuning using only response data can still yield strong instruction-following capabilities. Furthermore, GRAPE [9] focuses on response-specific valuation, revealing that the alignment between responses and the model’s internal knowledge significantly affects fine-tuning performance. These findings highlight the structural differences between instructions and responses.

This structural asymmetry, largely ignored by existing valuation approaches, has created a critical “valuation blind spot” in the post-training phase—obscuring the true drivers behind performance improvements. Treating instruction-response pairs as monolithic units, or focusing on only one component in isolation, discards important structural information and limits the potential for accurate attribution and effective data selection.

To close this gap, we will build a structure-aware data-valuation framework tailored to post-training. Our plan is to: 1) Conduct controlled perturbation and reconstruction experiments to quantitatively analyze the marginal contributions of instructions and responses and their impact on fine-tuning outcomes; 2) Establish a composite valuation framework that leverages structural information both within and across samples to reduce redundancy, capture complementarity, and enhance dataset representativeness and diversity; 3) Develop efficient, retraining-free evaluation methodologies to support transparent and trustworthy data selection processes.

5.2 Research on Structure-Aware Post-Training Data Valuation

Building upon the foundation established in Section 5.1, this project aims to develop a structure-aware data valuation framework to enhance the efficiency and interpretability of post-training processes for LLMs. Although instruction-response pairs serve as the fundamental units in instruction tuning, most existing methods overlook their inherent structural asymmetry and treat them as inseparable wholes, thereby limiting the precision of data attribution and the quality of data selection. While prior studies have partially revealed the functional differences between instructions and responses [5], such distinctions have yet to be systematically quantified or effectively leveraged in valuation methodologies. To fill this gap, we propose a general framework adaptable to various fine-tuning scenarios, comprising: 1) the design of controlled experiments to quantify the marginal contributions of instructions and responses; 2) the development of a composite valuation mechanism that accounts for both intra-sample structural roles and inter-sample interactions; 3) the implementation of retraining-free and interpretable evaluation methods to ensure transparency and reliability in data selection processes.

5.2.1 Subtask on Quantifying Structural Asymmetry Between Instructions and Responses

Our preliminary analysis shows that treating instruction-response pairs as indivisible units overlooks the structural asymmetry between them and reduces the granularity of data valuation. The instruction primarily provides task definition and contextual framing, while the response carries the actual solution and

knowledge. This functional difference may cause the model to react unequally to perturbations in each part. To reveal this structural asymmetry, we seek to design a perturbation-based evaluation framework that independently assesses the marginal impact of instructions and responses on model performance.

Specifically, we consider two types of perturbation strategies: Hard Noise and Soft Noise. Hard noise is injected at the token level by adding spelling errors, irrelevant words, or logical contradictions while preserving general semantic coherence. The perturbation intensity is controlled by word error rate (WER) to ensure consistency. Soft noise is applied at the embedding level by perturbing the input representations with Gaussian or mean-based noise. Inspired by Neftune [10], we implement the following embedding-level noise formulation: $X'_{\text{emb}} \leftarrow X_{\text{emb}} + \left(\frac{\sqrt{\alpha}}{Ld}\right) \epsilon$, where $\epsilon \sim \text{Uniform}(-1, 1)$ is sampled noise, with L being the sequence length and d the embedding dimension.

To assess the structural sensitivity, we perform fine-tuning on perturbed datasets derived from Alpaca, Dolly, and WizardLM, and evaluate performance across instruction-following and task-solving benchmarks. We define the marginal impact score as the absolute performance drop after applying perturbations on instructions or responses: $\Delta_I = |\text{Perf}(D) - \text{Perf}(D_{\text{perturb-instr}})|$, $\Delta_R = |\text{Perf}(D) - \text{Perf}(D_{\text{perturb-resp}})|$, where D denotes the original dataset, $D_{\text{perturb-instr}}$ and $D_{\text{perturb-resp}}$ refer to datasets with perturbations applied only to instructions or responses, respectively.

Based on this, we define the *Structural Sensitivity Ratio (SSR)* to measure the relative contribution of instruction versus response:

$$\text{SSR} = \frac{\Delta_I}{\Delta_I + \Delta_R} \quad (1)$$

A higher SSR ($\rightarrow 1$) indicates the model is more sensitive to instruction perturbations, implying higher importance of task formulation. Conversely, a lower SSR ($\rightarrow 0$) suggests that response content is more critical in model adaptation.

Although perturbation-based experiments offer a direct estimate of sensitivity, they may introduce unnatural noise artifacts. To complement this, we introduce a model-driven structural reconstruction strategy using large language models (e.g., GPT-4 or Claude) to generate semantically equivalent rewrites for both instructions and responses. These rewrites differ in syntax, phrasing, and complexity, allowing us to explore the model’s preference over natural structural variations.

We organize the rewrites into a *Structural Reconstruction Graph*, where each node represents a rewritten variant, and edge weights denote semantic similarity. By observing performance variations across this graph during training, we identify structurally sensitive regions in the instruction and response space.

In this task, our design will be focused on the following two issues: 1) design a perturbation-based metric suite that can reliably quantify functional asymmetry between instructions and responses, and 2) build a model-guided rewriting mechanism to generate natural structural variants for deeper analysis of model sensitivity patterns.

5.2.2 Subtask on Designing a Structure-Aware Composite Valuator

Based on the preceding analysis of structural heterogeneity, this task introduces the *Structure-aware Composite Valuator (SCV)*, which globally measures three dimensions of value: the robustness-driven generalization value of the instruction, the key-information value of the response, and the structural matching value produced by their combination. The rationale is three-fold: 1) highly robust instructions act as “anchors” of task intent and delimit the model’s generalization boundary for diverse inputs; 2) highly sensitive responses often contain irreplaceable core information [4]; 3) even when I and R are individually strong, a semantic mismatch can still impair training, so an explicit interaction score is required.

Instruction/Response scores. For each sample (I, R) we inject small perturbations δ_I, δ_R in the embedding space (Gaussian or mean noise, scaled by the Neftune rule $\sqrt{\alpha}/Ld$), create $(I + \delta_I, R)$ and $(I, R + \delta_R)$, compute $D_I = \text{KL}(P(I, R) \parallel P(I + \delta_I, R))$ and $D_R = \text{KL}(P(I, R) \parallel P(I, R + \delta_R))$, and set $S_I = \exp(-\mathbb{E}_{\delta_I}[D_I])$ (instruction robustness) and $S_R = \mathbb{E}_{\delta_R}[D_R]$ (response information scarcity).

Interaction score. To capture the semantic and structural compatibility between I and R , we adopt the V-Information formulation: $S_{\text{int}} = H_V(R) - H_V(R \mid I)$, which quantifies the predictive information gain about the response when the instruction is available. Here, $H_V(R)$ refers to the *predictive V-entropy*—the expected uncertainty of generating R without any instruction context. It reflects the inherent unpredictabil-

ity or ambiguity of the response alone. In contrast, $H_V(R | I)$ is the *conditional V-entropy*, which measures the uncertainty of predicting R when the instruction I is provided as context. If the instruction and response are well aligned, this conditional entropy should be substantially lower than the unconditional one. Thus, a higher difference S_{int} implies stronger coupling between I and R , indicating that the instruction provides meaningful guidance for generating the response.

Adaptive weight fusion. Employing the structural sensitivity prior $\text{SSR} = \Delta_I / (\Delta_I + \Delta_R)$ obtained in the previous task, we set $\alpha = \text{SSR}$, $\beta = 1 - \text{SSR}$, and $\gamma = \eta \min(\alpha, \beta)$ with default $\eta = 0.5$. The composite valuation is $\text{Value}(I, R) = \alpha S_I + \beta S_R + \gamma S_{\text{int}}$. Thus, if the model is more sensitive to instructions, α dominates; if more sensitive to responses, β dominates; and γ follows the smaller of the two, ensuring the interaction term is amplified only when both parts are valuable.

Diversity adjustment. To mitigate structural homogeneity, we encode instructions and responses as $\mathbf{z}_I = \phi_{\text{instr}}(I)$ and $\mathbf{z}_R = \phi_{\text{resp}}(R)$, compute the weighted distance $D_{ij} = \lambda \|\mathbf{z}_I^{(i)} - \mathbf{z}_I^{(j)}\|_2^2 + (1 - \lambda) \|\mathbf{z}_R^{(i)} - \mathbf{z}_R^{(j)}\|_2^2$, and estimate local density via a kernel $\rho(x)$. For low-density samples we multiply Value by $(1 + \kappa/\rho)$ (with smoothing constant κ) to prioritise sparse yet high-quality cases. Finally, we rank by the adjusted Value and, under budget N , select the top- K subset, achieving a triple trade-off among independent valuation, structural matching, and global diversity.

5.2.3 Subtask on Efficient and Explainable Evaluation

In the previous tasks, we aim to build the structural sensitivity measurement mechanism and the composite valuation model between instructions and responses, which providing foundational support for evaluating the quality of data subsets. However, existing methods still face two core challenges in practical: 1) Strategy validation heavily relies on complete retraining, which is computationally expensive and cannot support high-frequency iterations; 2) The sample selection process lacks clear causal attribution, making it difficult to explain why a particular data instance is selected. To address these issues, we design a structure-aware rapid strategy evaluation and causal attribution framework to enable efficient, transparent, and trustworthy data selection.

The core idea is to transform the strategy evaluation problem into a *sparse rating prediction* task in Collaborative Filtering (CF)—by treating historical subset-performance pairs as "user-item-rating" triplets, we achieve matrix completion for unknown subset effects through low-rank decomposition and structural feature regularization.

Collaborative Filtering-Based Performance Prediction. Let $\mathcal{P} = \{1, \dots, P\}$ be the set of fine-tuning strategies with executed experiments, and $\mathcal{F} = \{1, \dots, F\}$ be the set of structural features. To ensure the proxy model is both discriminative and concise, we retain only four groups of core features most relevant to final performance, with the total dimensionality controlled within twenty: The first group captures *central tendencies*, including instruction robustness mean μ_I , response information scarcity mean μ_R , and interaction coupling mean μ_{int} ; the second group measures *dispersion*, taking the standard deviations $\sigma_I, \sigma_R, \sigma_{\text{int}}$ of the above three metrics to characterize subset heterogeneity; the third group evaluates *structural diversity* using the average structural distance $\bar{D}$ and the inverse of local density mean $1/\bar{\rho}$, jointly measuring redundancy and coverage; finally, we include *scale indicators*—subset proportion $\frac{|\mathcal{S}|}{|\mathcal{X}|}$ and task label coverage cov_{task} . These features can be quickly computed using existing statistics from the first two subtasks.

For each strategy p , we discretize its selected subset's structural feature vector $\mathbf{x}_p$ into binary indicators $\mathbf{q}_{pf} \in \{0, 1\}$ (e.g., "high robustness interval" or "low density interval"), obtaining the strategy-feature matrix $Q \in \mathbb{R}^{P \times F}$. The actual fine-tuning performance gain r_p is recorded. We construct a *latent semantic factor model*: Let $U \in \mathbb{R}^{P \times k}$ and $V \in \mathbb{R}^{F \times k}$ represent strategy and feature latent vectors, respectively, with the predicted gain $\hat{r}_p = \sum_f Q_{pf} U_p^\top V_f$. The training objective is:

$$\min_{U, V} \sum_{p=1}^P (r_p - \hat{r}_p)^2 + \lambda (\|U\|_F^2 + \|V\|_F^2) + \mu \sum_f \frac{1}{\sigma_f} \|V_f\|_2^2, \quad (2)$$

where σ_f quantifies feature informativeness. The *structure-aware regularization* term $\frac{1}{\sigma_f}$ imposes smaller penalties on high-information features, allowing greater expressiveness. The learned V latent vectors assign *collaborative effect weights* to structural attributes, while U captures strategy preferences in the latent space. For a new strategy $\mathcal{S}$, we only need to compute its structural indicators $\mathbf{q}_{\mathcal{S}}$ and predict:

$$\hat{r}_S = \sum_f q_{Sf} \bar{U}^\top V_f, \quad \text{where } \bar{U} = \frac{1}{P} \sum_p U_p. \quad (3)$$

Here, $\bar{U}$ serves as the mean latent vector for cold-start strategy approximation, avoiding retraining of the main model.

Explanation Mechanism: Structure-Aware Shapley Attribution. Beyond performance prediction, we adopt the Shapley value [11] to explain individual sample contributions to model gains. Let the fine-tuned performance function be $v(S)$, where S is a sample set. Using the collaborative filtering model, we approximate $v(S)$ in a separable latent factor form: $v(S) = \sum_f (\sum_{(I,R) \in S} q_{(I,R)f}) w_f$. The structural Shapley value for a sample (I, R) is then: $\phi(I, R) = \frac{1}{|T|} \sum_{S \subseteq T \setminus \{(I,R)\}} \frac{1}{\binom{|T|-1}{|S|}} [v(S \cup \{(I,R)\}) - v(S)]$. Under the latent factor decomposition assumption, a greedy approximation is used: Select $k \ll |T|$ representative subsets S_k and compute: $\tilde{\phi}(I, R) = \sum_f q_{(I,R)f} w_f - \frac{1}{k} \sum_{(I',R') \in S_k} \sum_f q_{(I',R')f} w_f$. This approximation preserves Shapley’s efficiency and fairness while explicitly decomposing the structural contributions of instructions and responses, enabling interpretable evaluation.

6 Requested Funding and Budget

The total funding requested is \$ 250,000 HKD, which is broken down as follows:

Budget item	Detail breakdown and justification	Estimate cost
Staffing	Recruiting one PhD student Justification: $17,510 \text{ HKD} \times 12 \text{ months} \times 1 \text{ persons} = 210,120 \text{ HKD}$	210,120 HKD
Conference	The research outcomes of this project are expected to be published at top-tier conferences, e.g., ACM SIGKDD.	20,000 HKD
Total		230,120 HKD

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